

ITEMS		CONFIRM
1	<u>GENERAL</u>	
1.1	Scope	
1.1.1	This general specification covers the minimum requirements for the design, materials, fabrication, inspection and testing of trays.	
1.1.2	The trays include all the component parts of trays such as tray panels, weirs, downcomers, troughs, baffles, risers, caps, valves, support beams, clamps, bolts, gaskets and other specialties of proprietary trays. The support rings, downcomer bars and support brackets welded in the shell are out of the supply scope of tray vendor.	
2	<u>DESIGN</u>	
2.1	Materials for trays shall conform to those specified on the respective Tray Specification Sheet.	☐ YES ☐ NO
2.2	Loads Deflections and Allowable Stress	
2.2.1	Trays, pans, draw-off boxes, or similar internals shall be designed using a corroded thickness to support their own weight plus live load or maintenance load as	
1)	Design live load for fractionating trays shall be the greater of 150 kg/m ² or the weight of water 150 mm over the active tray surface.	☐ YES ☐ NO
2)	Design live load for areas under downcomers shall be the greater of 300 kg/m ² or the weight of water filled two-thirds of the downcomer.	☐ YES ☐ NO
3)	Design live load for pans (e.g. draw-off or accumulator) shall be the greater of 150 kg/m ² or the weight of water at the maximum operating level of the pan.	☐ YES ☐ NO
4)	Concentrated load at ambient temperature for tray inspection and installation shall be 140 kg, at any point.	☐ YES ☐ NO
2.2.3	Maximum allowable deflection of tray assemblies shall be <u>3 mm for small-diameter trays (up to 2,400 mm)</u> , and <u>1/900 of the shell diameter for larger trays</u> . Deflections shall be determined in both the un-corroded and corroded states.	☐ YES ☐ NO
2.2.4	Allowable bending stress of strength members of the tray shall be determined as follows:	☐ YES ☐ NO
1)	For the design live load:	
	$S_b = S_y @ T_{design} \times 1 / 1.6$ if $2.5 S_y \leq S_t$	
	$S_b = S_t @ T_{design} \times 1 / 4$ if $2 S_y \leq S_t \leq 2.5 S_y$	
	$S_b = S_y @ T_{design} \times 1 / 2$ if $S_t \leq 2 S_y$	
	where, S_b Maximum allowable bending stress	
	$S_t @ T_{design}$ Tensile strength at design temperature	
	$S_y @ T_{design}$ Yield strength or 0.2% offset yield strength at design temperature, whichever is smaller.	
2)	For short time maintenance load:	
	$S_b = \text{minimum of } (S_t @ T_{ambient} \times 1 / 1.5 , S_y @ T_{ambient} \times 90\%)$	
	where, $S_t @ T_{ambient}$ Tensile strength at ambient temperature (35°C)	
	$S_y @ T_{ambient}$ Yield strength or 0.2% offset yield strength at ambient temperature, whichever is smaller.	
	* Refer to ASME Code, Section VIII	

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2.2.5	Tray supports shall prevent trays from being lifted by upflowing vapor. Trays shall be designed to withstand an uplift pressure of three times the tray pressure drop at maximum vapor and liquid loads or 0.02 kg/cm ² , whichever is greater.	☐ YES ☐ NO																														
2.3	Dimensional Requirements																															
2.3.1	All parts of trays shall be accessible for inspection, cleaning and repair through the internal manways without the need of removing an excessive member of tray components.	☐ YES ☐ NO																														
2.3.2	Parts shall be designed and fabricated so that corresponding parts are interchangeable .	☐ YES ☐ NO																														
2.3.3	Tray panels shall be small enough to be removable through the column manholes and to avoid vibration in trays. 1) Thickness of tray panels shall not be less than 3 mm 2) Width of tray panels shall not exceed 380 mm 3) Length of tray panels shall not be longer than 2,000 mm	☐ YES ☐ NO ☐ YES ☐ NO ☐ YES ☐ NO																														
2.3.4	All parts except tray manways shall be designed to permit installation and removal from the top side of trays.	☐ YES ☐ NO																														
2.3.5	The minimum overlap between tray floor sections and support members shall be 20 mm.	☐ YES ☐ NO																														
2.3.6	Fastening of downcomer panels to the vertical downcomer support bars shall permit some on-site adjustment for ensuring correct setting of clearances under downcomer.	☐ YES ☐ NO																														
2.3.7	Spacing of bolts or clamps shall be close enough to insure optimum liquid-tight construction. Bolt hole spacing around the edge of tray panels shall not exceed 175 mm. Bolt hole spacing in downcomers, pans or bubble cap trays shall not exceed	☐ YES ☐ NO																														
2.3.8	Standard width of tray support rings and downcomer clamping bars shall be as follows: <table><tr><th rowspan="2">Tower Diameter</th><th rowspan="2">Support Ring mm</th><th colspan="2">Downcomer Bar</th></tr><tr><th>Side mm</th><th>Center mm</th></tr><tr><td>Up to 900</td><td>40</td><td>75</td><td>--</td></tr><tr><td>Over 900 to 2,100</td><td>50</td><td>75</td><td>125</td></tr><tr><td>Over 2,100 to 3,900</td><td>65</td><td>90</td><td>150</td></tr><tr><td>Over 3,900 to 6,000</td><td>75</td><td>125</td><td>150</td></tr><tr><td>Over 6,000 to 7,500</td><td>90</td><td>125</td><td>165</td></tr><tr><td>Over 7,500</td><td colspan="3">require individual design</td></tr></table>	Tower Diameter	Support Ring mm	Downcomer Bar		Side mm	Center mm	Up to 900	40	75	--	Over 900 to 2,100	50	75	125	Over 2,100 to 3,900	65	90	150	Over 3,900 to 6,000	75	125	150	Over 6,000 to 7,500	90	125	165	Over 7,500	require individual design			☐ YES ☐ NO
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2.3.9	To avoid thermal expansion problems, Vendor shall provide adequate fastener spacing or slotted holes, and establish satisfactory clearances between the trays and the shell, [13 mm per 3,000 mm]	☐ YES ☐ NO																														
2.3.10	Tray holes shall be avoided in that part of the tray located over the support rings .	☐ YES ☐ NO																														
2.4	Corrosion Allowance Corrosion allowance is specified in the Specification Sheets.	☐ YES ☐ NO																														

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2.5	Minimum Thickness Thickness of tray parts is specified on Specification Sheet. Minimum tray thickness is 2 mm (14 gage) for corrosion-resistant alloys and 3.4 mm (10 gage) for carbon steel.	☐ YES ☐ NO
2.6	Cartridge Tray Cartridge type trays shall be used for small columns less than 800 mm in diameter.	☐ YES ☐ NO
2.6.1	Cartridge type trays shall be designed to minimize leakage from the periphery.	☐ YES ☐ NO
2.6.2	Cartridge trays may be removable from the top only.	☐ YES ☐ NO
2.7	Downcomers Stiffeners must be provided as needed to limit the horizontal movement of the bottom edge of the downcomer at any point to 6 mm under liquid head pressure.	☐ YES ☐ NO
2.7.1		☐ YES ☐ NO
2.8	Seal Pans, Draw-off Boxes All joints in draw-off boxes, or seal pans shall be of liquid tight construction by using gaskets or seal welded, unless otherwise specified.	☐ YES ☐ NO
2.9	Tray Manways Access through trays shall be provided as follows:	☐ YES ☐ NO
2.9.1	1) All trays of 1200 mm and larger in diameter shall be equipped with manways. 2) All trays smaller than 1200 mm in diameter shall be either split or equipped with manways	☐ YES ☐ NO
2.9.2	Tray manways shall be provided as follows:	☐ YES ☐ NO
	1) Manway Size shall be at least 350 mm x 400 mm, but not longer than 500 mm.	☐ YES ☐ NO
	2) Mayway panel weight shall not exceed 25 kg.	☐ YES ☐ NO
	3) Manway shall be top- and bottom-removal.	☐ YES ☐ NO
	4) When major support beams are required, each side shall have one manway because those hinder passage from one side to the other.	☐ YES ☐ NO
	5) For multipass trays, one tray manway is required for each tray pass because the central downcomers restrict access from one side to the other.	☐ YES ☐ NO
	6) Partial vertical alignment of manways shall be provided for easy passage of lights, hoses, and to allow personnel to stand erect. Full alignment through whole column is not allowable because it increases the distance that personnel or tools can fall freely down the column.	☐ YES ☐ NO
	7) Tray manways shall not form a part of the downcomer seal area.	☐ YES ☐ NO

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2.10 Support Beam	
2.10.1 Support beams should carry the highest likely distributed load on the trays without exceeding the maximum allowable deflection as described in item 2.2.3. 1) In large columns ($D > 2,700 \text{ mm}$), one or more major support beams are required, and these also support the minor support beams. 2) In smaller columns, major support beams are not required, and trays are entirely supported by the support rings and minor support beams.	☐ YES ☐ NO
2.10.2 Support beams shall be located as follows: 1) In large columns ($D > 2,700 \text{ mm}$), the major support beams shall be installed <u>parallel</u> to the liquid flow, and the minor beams perpendicular to the liquid flow. 2) In smaller columns, support beams shall be <u>parallel</u> to the liquid flow on the trays in order to minimize vapor crossflow.	☐ YES ☐ NO
2.10.3 Vertical height of beams perpendicular to the liquid flow shall not exceed 20 percent of the tray spacing, or 75 mm, or 5 percent of the column diameter, whichever is smaller.	☐ YES ☐ NO
2.10.4 Width of beams shall not exceed 75 mm	☐ YES ☐ NO
2.10.5 Slotted holes shall be used on the joints of minor beams to the major beam to allow leveling.	☐ YES ☐ NO
2.10.6 Slotted holes with sufficient clearance for thermal expansion shall be used where support beams are bolted to clips.	☐ YES ☐ NO
2.11 Fastening of Trays, Downcomers, and Support Beams	
2.11.1 Tray panels shall be fastened to the support rings and support beams with frictional type tray clamps or bolts. When tray vibration is expected, bolted-through tray connections shall be used, along with lock washers or self-locking nuts, rather than frictional fastening.	☐ YES ☐ NO
2.11.2 Bolts or clamps shall be accessible from the top side of the trays for easier removal.	☐ YES ☐ NO
2.11.3 Clamping shall be used for installation of trays on the support rings.	☐ YES ☐ NO
2.11.4 Minimum diameter of bolting shall be 10 mm (M10). Larger bolts shall be required for major support beams.	☐ YES ☐ NO
2.11.5 All bolting shall be furnished in the Metric system of units.	☐ YES ☐ NO
2.11.6 Gasket and packing shall not be used in the design or installation of trays except when specified. Fabrication shall provide tight metal-to-metal joints. Metal seal plates may be used to close construction joints where necessary.	☐ YES ☐ NO
2.12 Tray Drainage	
2.12.1 Trays and seal pans not specified to be liquid-tight and of a design which would not easily drain, shall be provided with one or more 6 to 12 mm diameter weep holes to permit draining of the liquid hold-up.	☐ YES ☐ NO
2.12.2 Weep holes in recessed seal pans shall be located downstream or on the tray side of the downcomer so that any vapor passing through the holes will not by-pass the tray. Inlet weirs shall have a 6 to 12 mm hole at the base to drain into the tray.	☐ YES ☐ NO
2.12.3 When weep holes are provided in bubble-cap or valve trays, they shall be located as close to the outlet weir as possible.	☐ YES ☐ NO
2.12.4 All boxes and pans from which liquid is normally withdrawn shall be equipped with drain nozzle that will insure complete drainage of the box and pan. Weep hole shall not be provided in the water draw-off boxes.	☐ YES ☐ NO

sheet 5 of 7

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3	<u>FABRICATION</u>	
3.1	Forming	
3.1.1	Bending shall be by cold press-bending.	☐ YES ☐ NO
3.1.2	Press-forming shall be made in such a manner that wrinkling and roughening may not be caused.	☐ YES ☐ NO
3.1.3	Holes on tray panels shall be so punched as to minimize burring and shall be punched from upper side .	☐ YES ☐ NO
3.1.4	<u>Nuts tack-welded to tray bottom</u> shall be applied by tray vendor.	☐ YES ☐ NO
3.2	Surface Finishing	
3.2.1	Surfaces of fabricated parts shall be smooth and free from dents, hammer marks, kinks or other defects which particularly might prevent close metal-to-metal fit.	☐ YES ☐ NO
3.2.2	Burrs shall be removed from perforated areas and edges of tray sections.	☐ YES ☐ NO
3.2.3	Welds shall be smooth and free from slag and spatter.	☐ YES ☐ NO
3.2.4	Welded portion for stainless steel shall be pickled.	☐ YES ☐ NO
3.3	Welding	
3.3.1	Welding shall be done by a metal-arc process.	☐ YES ☐ NO
	Resistance welding is permitted for minor details on approval of the Purchaser.	
3.3.2	Welding electrodes shall be in accordance with the respective standards of ASTM or JIS.	☐ YES ☐ NO
3.3.3	Tray support rings and bars shall be continuously welded on the top and side and stitch welded on the bottom. Brackets shall be continuously welded on the top and sides only.	☐ YES ☐ NO
3.3.4	Parts to be welded shall be thoroughly cleaned free from moisture, oil, grease, paint, scale and other foreign materials prior to welding.	☐ YES ☐ NO
3.3.5	When trays and beams are welded, jigs and other positioning devices shall be used to minimize deformation.	☐ YES ☐ NO
3.4	Marking	
3.4.1	Component parts of assemblies shall be permanently marked with piece marks corresponding to those shown on the vendor's detailed and assembly drawings.	☐ YES ☐ NO
3.4.2	Interchangeable pieces shall have the same mark.	☐ YES ☐ NO
3.5	Fabrication Tolerance	
3.5.1	Height of risers, and baffles, and clearances between caps and panels shall be fabricated within 1.5 mm of the dimensions specified, and the lengths of downcomers shall be within 3 mm.	☐ YES ☐ NO
3.5.2	Height of weirs shall be fabricated within 1.5 mm of the dimensions specified.	☐ YES ☐ NO
3.5.3	Clearances between bottom edge of downcomer and tray panels shall be within 3 mm of the dimensions specified.	☐ YES ☐ NO
3.5.4	Dimensions from weirs to shell plate of column shall be within 9 mm of the dimensions specified.	☐ YES ☐ NO
3.5.5	Hole diameters of sieve trays shall be within 1% of the diameters specified. Number of holes shall be within plus or minus 2% of the number specified.	☐ YES ☐ NO
3.5.6	Dimension tolerances for proprietary trays, such as Nutter Float Valve, shall be in accordance with tray vendor's standard.	☐ YES ☐ NO
3.5.7	All other parts shall be fabricated in accordance with good shop practice and in such uniformity that all the corresponding parts will be interchangeable.	☐ YES ☐ NO

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4	<u>INSPECTION AND TESTS</u>	
4.1	General	
4.2	Inspection of Materials	
4.2.1	All the materials rejected by the Purchaser, by virtue of defects or incorrect material, shall be replaced promptly by the vendor.	☐ YES ☐ NO
4.2.2	Mill certificate and Mill test report for alloy steel and non ferrous shall be submitted.	☐ YES ☐ NO
4.3	Visual Inspection Visual inspection of tray parts shall be carried out for the following items: 1) Surface finishing 2) Surface cleaning 3) Appearance 4) Burring of punched hole 5) Harmful deformation	☐ YES ☐ NO
4.4	Testing of Assembly	
4.4.1	At least one of each type of tray shall be trial assembled in the supplier shop using parts selected at random. The trial assembled shall be supported in a manner comparable to the final installation.	☐ YES ☐ NO
4.4.2	The maximum tolerance for shop assemblies with respect to a flat plane shall be ± 3 mm.	☐ YES ☐ NO
4.4.3	The tolerance for field assemblies with respect to a level plane shall be ± 3 mm for column diameters through 4,500 mm and ± 5 mm for column diameters greater than 4,500 mm.	☐ YES ☐ NO
4.4.5	The final levelness check during tray installation is to be made using an optical level. Where there are focusing restrictions, a manometer may be used for levelness checking.	☐ YES ☐ NO
4.5	Leakage Test for Bubble-cap Trays Hydraulic leakage test shall be carried out in the manufacturer's shop: Fill the tray with water (free of chloride) up to the level of outlet weir and check for drop in water level, which shall be not more than 1 mm in 20 min.	☐ YES ☐ NO
5	<u>PREPARATION FOR SHIPMENT</u>	
5.1	General	
5.1.1	All the carbon steel surface of trays shall be coated with a suitable rust preventive prior to shipment. Rust preventive method shall be submitted to the Purchaser.	
5.1.2	All internal parts inclusive of trays shall be clearly identified with corresponding marks provided on the assembly drawings to permit rapid assembly. No paint shall be used for marking.	☐ YES ☐ NO

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5.2 Packing 5.2.1 Tray parts shall be properly protected to prevent deformations, damage and rusting. 5.2.2 Tray parts for different columns shall not be intermixed. 5.2.3 Small items shall be properly packed to prevent loss in transit and detailed packing lists shall be attached to packages.	☐ YES ☐ NO ☐ YES ☐ NO ☐ YES ☐ NO
6 <u>DRAWINGS AND INFORMATION</u> 6.1 Vendor shall provide the drawings as follows: 1) Installation Assembly of Tray and Standard Hardware for the project. 2) Assembly of Tray 3) Detail of Tray 4) Detail of Tray Support and Seal Pan 5) Detail of All other Internals 6.2 Vendor shall provide the following calculation sheets . 1) Hydraulic calculation sheet for trays 2) Loads deflection and stress in items 2.2.1 through 2.2.5 6.3 All drawings and documents shall be prepared on the following condition: 1) Measurement system: Metric system 2) Language: English or Chinese	☐ YES ☐ NO ☐ YES ☐ NO ☐ YES ☐ NO ☐ YES ☐ NO ☐ YES ☐ NO ☐ YES ☐ NO ☐ YES ☐ NO ☐ YES ☐ NO
7 <u>GUARANTEE</u> For proprietary tray and liquid distributor, Vendor guarantees that the performance of tray should be consistent with the calculation sheet in quotation.	☐ YES ☐ NO
8 <u>SPARE PARTS</u> 8.1 Sieve trays: The vendor shall provide the spare parts including bolts, nuts, washers and clamps, as follows: 1) For Stainless Steel: 3% excess 2) For Hastelloy C: 5% excess 8.2 Bubble cap trays: 1) Bubble Cap: 3% excess 2) Sealing Packing: 50% excess	☐ YES ☐ NO